

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1465

Brodhead Sw Sta 138 kV
December 16, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1465. The Interconnection Customer proposes a 237.3 MW solar photovoltaic facility interconnecting at the Brodhead Sw Sta 138 kV substation in the Virginia area. The request was received on April 23, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 433 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$440,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bateville - Innovation 138.0 kV Line	Medical - Eagleville 138.0 kV	110.0	279	20.4
Brodhead Sw Sta - Quantico 138.0 kV Line	Possum Point - Heiskell 138.0 kV	105.2	223	59.0
Bateville - Novec Country Clb 138.0 kV	Medical - Eagleville 138.0 kV	103.5	236	20.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bateville - Innovation 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,273,000	14% (\$174,000)
Brodhead Sw Sta - Quantico 138.0 kV Line	Reconductor line	\$192,000	48% (\$92,000)
Bateville - Novec Country Club 138.0 kV	Reconductor line	\$1,273,000	14% (\$174,000)
	Total allocated to project		\$440,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.